



OPEN Prediction of remaining useful life for electronic equipment based on online PINN

Feng Han^{1,2✉} & Bo Mo²

As critical components of spacecraft systems, electronic equipment requires accurate remaining useful life (RUL) prediction to ensure reliable operations. Traditional physics-based and data-driven methods are limited by poor generalization and substantial data demands, respectively. While Physics-Informed Neural Networks (PINNs) integrate physical laws to enhance accuracy, they remain susceptible to performance degradation under shifting operational conditions due to catastrophic forgetting. This paper proposes a novel online PINN framework integrated with continual learning to dynamically adapt to new degradation patterns. Validated on NASA's IGBT dataset, our method reduces RUL prediction error to 27.2% of a traditional LSTM model and achieves an online update time of less than 800ms/cycle. These results demonstrate a significant improvement in robustness and accuracy, providing a superior solution for RUL estimation of aerospace power electronics under extreme environments.

Keywords Aerospace, remaining useful life, Physics-Informed neural networks, Continual learning

The prognostics of remaining useful life (RUL) for spacecraft electronic equipment constitutes a critical technological foundation for ensuring mission success in aerospace engineering. As the central nervous system of modern spacecraft, onboard electronics operate under extreme orbital conditions characterized by intense radiation fluxes ($\pm 2 \times 10^{12}$ protons/cm²·s), drastic thermal cycling (-180 °C to + 150 °C), and prolonged zero-maintenance requirements (typically > 15 years)¹. The U.S. Defense Advanced Research Projects Agency (DARPA) reports that over 68% of spacecraft anomalies originate from electronic system degradation, with power electronics accounting for 42% of these failures². Current RUL prediction methodologies predominantly adopt two paradigms: physics-of-failure models based on accelerated life testing (ALT) and data-driven approaches utilizing telemetry analytics. The former leverages parametric degradation equations (e.g., modified Arrhenius-Eyring hybrid models³) but struggles with multi-physics coupling effects in actual space environments. The latter, empowered by machine learning architectures like CatBoost and DBSO-optimized networks, faces challenges in physical interpretability and operational condition adaptability. Figure 1 shows the real-time monitoring data of 8 key voltage channels on the integrated flight control power board, which not only reflects the dynamic response characteristics of the integrated flight control assembly under harsh working conditions, but also serves as an important basis for predicting the remaining life of the system. In the aviation industry, the gradual aging of the integrated flight control assembly can lead to increased power losses and rising operating temperatures, ultimately potentially causing catastrophic failures.

The Insulated Gate Bipolar Transistor (IGBT), as a high-performance power semi-conductor switching device, has found extensive applications in the aerospace sector due to its fast switching speed, high current-carrying capacity, and low switching losses. In spacecraft power systems, IGBTs play critical roles in solar array maximum power point tracking (MPPT) and lithium-ion battery charge-discharge management, where their high-voltage capabilities (up to 6.5 kV) ensure efficient energy conversion under extreme orbital conditions. For electric propulsion systems like ion thrusters, IGBT-based inverters enable kHz-level switching to generate precise high-voltage pulses for plasma acceleration, as demonstrated in NASA's Deep Space 1 mission. The device's radiation-hardened variants (e.g., SiC-IGBT hybrids in ESA's BepiColombo Mercury probe) address unique space challenges including cosmic ray-induced single-event effects and -150 °C to +150 °C thermal cycling. In attitude control subsystems, IGBT-driven brushless DC motors achieve micro-radian-level precision in reaction wheel assemblies, while commercial constellations like Starlink leverage their compact HiPak modules for high-density power distribution. However, the harsh space environment imposes stringent reliability requirements

¹Beijing Aerospace Automatic Control Institute, Beijing, China. ²School of Aerospace Engineering, Beijing Institution of Technology, Beijing, China. ✉email: 3120225045@bit.edu.cn

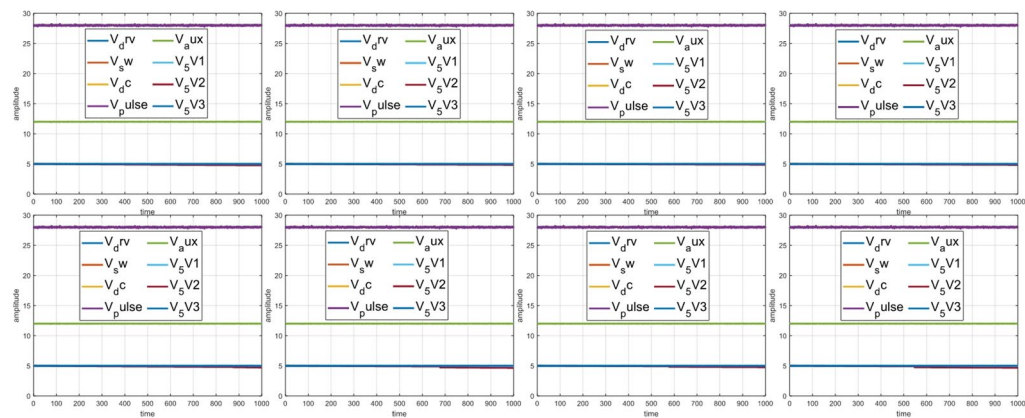


Fig. 1. Voltage output variation of Integrated flight control combination.

- typical spacecraft demand 15-year operational lifespans with zero maintenance, making RUL prediction paramount.

Currently, RUL estimation methods are mainly divided into two categories: physics-based prediction techniques and data-driven prediction techniques^{4–5}. Physics-based methods typically start from the internal working mechanisms of the relevant system, establishing mathematical models that reflect the physical laws of system performance degradation. For example, under the premise of unchanged storage conditions and failure mechanisms, Tao Xueqing and others initially analyzed the magnitude of stress applied during accelerated storage, considered the interaction of multiple load factors, and modified the Arrhenius and Eyring models to establish a model for converting product life under accelerated and natural conditions⁶. However, the obtained physical models are often not precise enough, and the modeling process can easily lose the nonlinear relationships between parameters, leading to increased errors.

Data-driven RUL assessment methods learn the mapping relationship between input and output from historical aging data of electronic device output characteristics. These methods are primarily divided into probability statistical methods and machine learning methods. Among them, machine learning methods first select several historical data as training samples and then use training algorithms to predict failures after training. Liu Jiacheng utilized NASA's publicly available aging experiment data for IGBT life prediction, analyzing and comparing the accuracy of model predictions⁷; Chen Changji et al., based on data feature extraction, employed the Differential Brainstorm Optimization (DBSO) algorithm to optimize the CatBoost model for fault diagnosis⁸. Machine learning methods avoid the complex establishment of traditional mathematical models and the acquisition of expert experience. However, they require complete historical data from the initial use to the final failure of the observed object; otherwise, the credibility of the prediction results may be reduced.

Physics-Informed Neural Networks (PINNs) integrate physical constraints into neural networks by embedding governing principles within the loss function⁹. This approach ensures that network training aligns with fundamental physical laws rather than relying purely on data-driven optimization^{10–11}, leading to more physically meaningful predictions. Initially introduced for solving mathematical equations such as ordinary and partial differential Eqs^{12–13}, PINNs have also been explored in power electronics for parameter estimation. Building upon parameter estimation, the application of PINNs has been extended to prognostics and health management in adjacent fields. However, their potential application in predicting the RUL of power electronic devices is still an area requiring further investigation.

Beyond physics-informed and continual learning approaches, recent studies have explored graph neural networks (GNNs) for fault detection and out-of-distribution recognition. For instance, the GCI-ODG framework leverages graph convolution and information propagation to detect complex industrial faults, while the EGN-OD framework employs energy-based graph neural networks to improve out-of-distribution fault detection performance. These methods demonstrate the versatility of GNN-based detection in capturing relational dependencies among signals. Although promising, they primarily focus on graph-based representations rather than integrating physics constraints and continual learning, which is the focus of this work.

However, a trained neural network model aligns with the degradation patterns found in historical data. When operational conditions shift, these patterns also change. Due to the nature of deep learning, retraining a life assessment model with new degradation data can degrade predictive performance^{14–15}, a challenge known as catastrophic forgetting. Additionally, gathering all past degradation data for retraining is costly and often impractical¹⁶. Continual learning enhances a model's ability to adapt to new tasks over time, mitigating catastrophic forgetting while maintaining learning efficiency.

Table 1 provides a comparative summary of related PINN-based approaches and highlights the distinctions of our proposed method.

Here are the innovative points of this paper:

Approach	Limitations	Advantages of Proposed Method
PINN + LSTM ¹⁷	Requires large amounts of training data; trained in an offline manner; fails to retain knowledge under changing conditions (catastrophic forgetting).	Introduces continual learning to achieve online adaptation and robustness against catastrophic forgetting.
PINN in RUL prediction ¹⁸	Improves physical consistency but still relies on offline training; performance degrades when operational conditions shift.	First framework to combine PINN with continual learning for online RUL prediction, ensuring both physical consistency and adaptability.

Table 1. Limitations of existing approaches and advantages of the proposed framework.

- (1) This study investigates PINNs for power electronics RUL estimation by integrating physical constraints as regularization terms within the loss function, which systematically combines data-driven learning with domain physics to enhance both prediction accuracy and physical consistency.
- (2) Combination of Physical Models and Data-Driven Approaches: This paper combines physical modeling with data-driven machine learning methods to overcome the limitations of using a single approach. By integrating models based on physical principles with statistical models derived from historical data, it achieves high accuracy in predicting the RUL.
- (3) Incorporation of Continual Learning Techniques: The paper discusses how deep learning models can remain effective under changing operational conditions by introducing continual learning mechanisms. This approach mitigates the issue of catastrophic forgetting, avoiding the high costs associated with retraining models using new degradation data.

In summary, to address the challenges of IGBT life assessment, this paper specifically proposes a new online learning strategy for RUL estimation of IGBTs and other power electronic devices. This strategy dynamically adjusts the model as operating conditions change, ensuring the accuracy and reliability of predictions. This advancement is vital for aerospace power control and propulsion systems, where predicting IGBT lifetime with high precision directly supports mission success and extends spacecraft operational life.

Related works
IGBT RUL

Model-based methods establish mathematical relationships between a device’s lifespan and its influencing factors. These models predict the failure cycle count (Nf) of IGBT devices, often attributing failure to plastic strain induced by thermomechanical mismatches in packaging layers¹⁹. The Coffin-Manson model, which focuses on temperature fluctuations (ΔT), provided the basis for later developments²⁰. The Nor-ris-Landzberg equation extends this by factoring in temperature cycling frequency (f)²¹. The Bayerer model integrates five physics-based parameters²²: peak junction temperature ($T(j, \text{max})$), heating duration (t_{on}), DC current (I), bond wire diameter (D), and blocking voltage (V). Additionally, cumulative damage models employing rainflow counting techniques are frequently used alongside analytical models for failure estimation²³. In addition to mainstream analytical models, there are also physical lifespan models based on energy hypotheses²⁴.

Neural networks (NNs) have been utilized for IGBT RUL prediction, demonstrating slightly better performance than comparable methods like Adaptive Neuro-Fuzzy Inference Systems (ANFIS). To enhance input quality, precursor signal data underwent preprocessing, with Principal Component Analysis applied to extract key time-domain features before being fed into feedforward neural networks. More advanced deep learning techniques have also been explored, including Long Short Term Memory (LSTM) networks, which have been employed for RUL estimation in IGBTs²⁵. While LSTMs offer improved accuracy over traditional model-based methods, they demand larger training datasets. Additionally, attention mechanisms have recently been introduced in IGBT RUL prediction for the first time.

PINN

PINNs have been applied in power electronics. Zhao et al. utilized PINNs for parameter estimation in DC-DC buck converters²⁶. Zhang et al. leveraged PINN-based physical modeling to assess the impedance of voltage source converters²⁷. Wu et al. introduced AutoPINN, integrating AutoML with PINNs for automated model design, demonstrating improved performance in parameter estimation for power electronic converters²⁸. Chen et al. developed the PI-LSTM method, combining PINNs with LSTMs to enhance parameter estimation in DC-DC buck converters²⁹.

Hybrid methods incorporate physical knowledge into data-driven modeling, combining the strengths of both model-based and data-driven approaches. This integration enhances predictive capabilities, making hybrid techniques a promising direction. Busca et al.³⁰ introduced a particle filter-based approach that embeds crack propagation physics to estimate the RUL of IGBT modules. Arias Chao et al.³¹ developed a hybrid frame-work inspired by PINNs, demonstrating improved RUL prediction for turbofan engines with reduced training data compared to purely data-driven methods. However, their approach did not tackle the challenge of catastrophic forgetting. More recently, Shen et al.³² proposed an embedded physical information network for blade crack detection, incorporating dynamic multi-level credibility into the training process to enhance interpretability and stability. While this study highlights the potential of embedding physics into deep learning for fault diagnosis, it remains limited to offline settings and does not address the continual adaptation problem that our work focuses on.

Continual learning

Several approaches have been developed to preserve past knowledge and mitigate catastrophic forgetting. Li et al. established knowledge distillation in continual learning through their Learning Without Forgetting (LWF) framework, enabling cross-task stability during sequential model training³³. This technique remains a cornerstone for continual learning, spawning diverse algorithmic extensions such as Dhar et al.'s attention distillation loss for preserving inter-task feature relationships³⁴. Douillard et al. introduced ensemble output distillation loss to limit feature changes³⁵. Rusu et al. created "Progressive Networks," which merge networks corresponding to different tasks into one³⁶. Abati et al. introduced task-specific gating modules within convolutional layers to selectively activate neurons for learning new tasks³⁷. However, such approaches introduce the stability-plasticity dilemma, where a continual learning system must balance retaining prior knowledge with adapting to new information.

Continual learning mimics human learning and aims to develop efficient models capable of continuous learning across a series of tasks. It mainly addresses two issues: solving the inherent problem of catastrophic forgetting in neural network design and improving the generalizability of training models. This ensures models have both plasticity (the ability to learn new knowledge) and stability (the ability to retain old knowledge), thus solving the stability-plasticity dilemma.

This work belongs to the hybrid methods for IGBT RUL estimation. It incorporates physical rules into network training and introduces continual learning to adapt to changing working environments, thereby advancing current data-driven NN methods. Recently, Shen et al.³⁸ proposed an unsupervised mixed-up differential replay mechanism with the concept of a "forgetting frontier" for blade crack propagation monitoring. Their work demonstrates the effectiveness of continual learning in mitigating catastrophic forgetting in industrial fault detection tasks. However, it remains focused on unsupervised anomaly detection rather than online RUL prediction, which is the main goal of our study.

Federated learning

Federated learning (FL) has emerged as a promising decentralized training paradigm for RUL prediction, as it enables collaborative model training across multiple clients while preserving data privacy. Xu et al.³⁹ proposed an adaptive sampling-based ensemble FL framework for aircraft engine RUL prediction, which dynamically adjusts client participation based on data quality. Their method effectively improves prediction accuracy and robustness under heterogeneous data distributions.

Building upon this line of research, several recent studies have further extended FL to address key challenges in industrial prognostics. Cai et al.⁴⁰ developed FedCov, a trustworthy FL framework that converts continuous RUL values into discrete categories to mitigate client heterogeneity. By incorporating a converter module and uncertainty estimation, FedCov alleviates model drift issues and achieves state-of-the-art performance on the C-MAPSS benchmark. Liu⁴¹ proposed a federated learning framework with prototype learning and physical information fusion for turbine blade RUL prediction. This approach integrates sensor monitoring data with digital twin-based physical degradation data, and leverages SHAP analysis to enhance model interpretability. Guo et al.⁴² introduced FedRUL, which adopts an edge-cloud collaboration mechanism combining local convolutional autoencoders (CAEs) with a global encoder and RUL predictor. This design addresses the computational and data limitations of edge clients while ensuring model confidentiality. Chen et al.⁴³ presented RUL-FLTNP, which integrates FL with Taylor-expansion-based network pruning to produce lightweight RUL predictors. Their method alternates pruning and recovery in local clients, enabling compact and efficient models suitable for deployment on edge devices.

These advances highlight the versatility of FL in addressing privacy, heterogeneity, interpretability, and deployment efficiency in decentralized RUL prediction. However, most existing FL-based methods remain focused on distributed learning and data privacy, without considering the issue of catastrophic forgetting in dynamically evolving operating environments. This limitation further motivates our proposed framework, which combines PINNs with continual learning to achieve online RUL prediction under non-stationary conditions.

Methodology

Considering the harsh operational conditions of space missions, where components are exposed to extreme thermal and mechanical stresses, accurate RUL prediction models are indispensable. Therefore, this paper utilizes a complex network model to handle degradation data and predict RUL. The entire process can be divided into several key steps and components, as shown in Fig. 2. First, the model receives real degradation data as input, representing the actual observed degradation of the equipment. Subsequently, the encoder processes the input data. The encoder's function is to compress and encode the input data, extracting important features known as feature values.

Once the feature extraction is complete, part of the feature values are transmitted to the decoder. The decoder's task is to reconstruct the predicted degradation data using these feature values. After generating the RUL predictions, these data are sent to the discriminator and the physical model. The discriminator compares the predicted results with the actual data, yielding a loss value to ensure the accuracy and validity of the data. The physical model uses actual physical laws and empirical predictions to validate the predicted degradation data, obtaining a physical model loss value. This ensures that the predictions are meaningful not only on a data level but also on a physical level.

Another part of the process involves using the network model to handle the feature values and generate RUL predictions, which are then compared to the actual RUL data. The network model adopts an encoder-decoder structure, extracting feature values from the input degradation data and generating predictions. These predictions are then analyzed and validated using the RUL prediction network. Finally, the accuracy and reliability of the predictions are ensured through the discriminator and the physical model. The model uses a

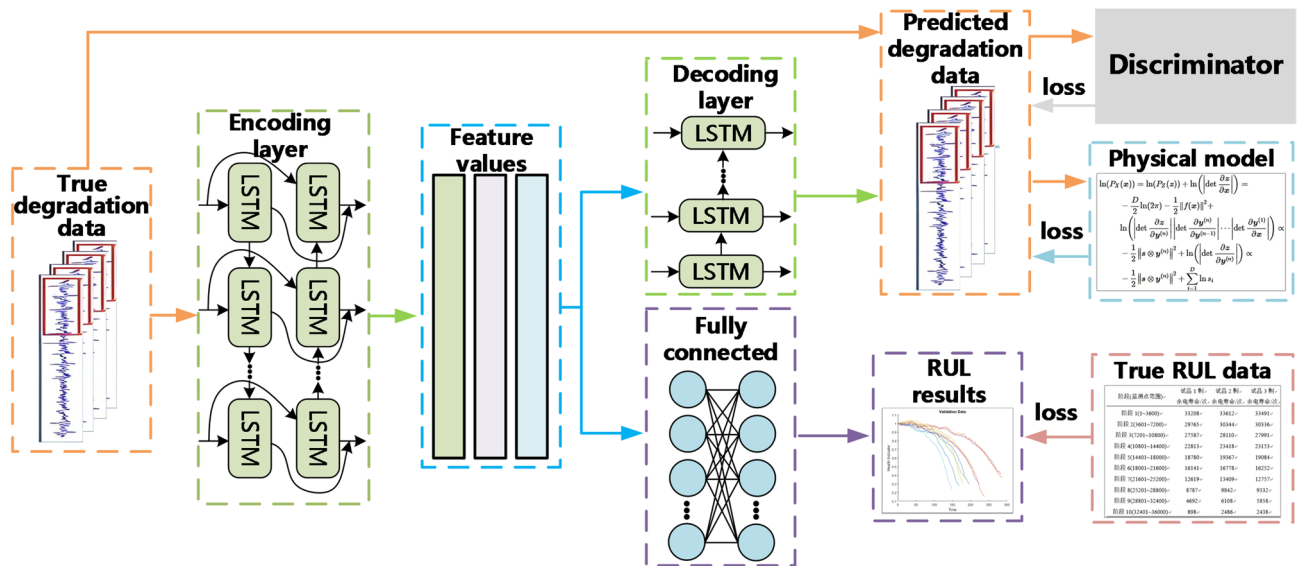


Fig. 2. Structure of networks.

3-layer BiLSTM encoder (256 units/layer) and a decoder with LSTM + 2 dense layers (128→64 units), integrating multi-head attention (8 heads), and the Dropout rate is 0.3.

Framework overview

Our proposed online PINN framework integrates four core components to address the challenges of RUL prediction under dynamic operational conditions: (1) an attention-based LSTM encoder-decoder network for temporal feature extraction and RUL prediction, (2) a discriminator module for ensuring prediction quality through adversarial training, (3) physics-informed regularization incorporating semiconductor degradation constraints, and (4) a continual learning mechanism enabling online adaptation without catastrophic forgetting.

The framework operates in two phases: an initial offline training phase using historical degradation data, followed by an online adaptation phase where the model continuously updates as new operational data becomes available. The attention-based LSTM processes time-series degradation signals and generates initial RUL predictions. The discriminator validates prediction quality by distinguishing between real and predicted degradation patterns. Physical constraints ensure predictions conform to semiconductor failure physics through monotonicity and boundary conditions. Finally, continual learning enables dynamic adaptation while preserving historical knowledge through the DAARN architecture.

The integration of these components creates a unified optimization objective that balances data-driven accuracy, physical consistency, and adaptive capability, specifically designed for the demanding requirements of aerospace power electronics monitoring.

Attention-based LSTM network

This paper enhances LSTM models with an attention mechanism to improve representation. The LSTM encoder processes input sequentially, generating hidden states. The attention mechanism assigns weights to these states, highlighting key time steps. The weighted sum then forms the final representation vector for prediction. The formula is as follows:

$$G = \sum_i^k s_i g_i \quad (1)$$

Where G represents the result of attention calculation, denotes the computed weight value, and represents collected historical wind speed data.

The LSTM model captures long-term dependencies by updating hidden states sequentially, where each state builds upon the previous one and carries information forward. Context is preserved through hidden states, enabling the use of prior information for accurate predictions. After applying the attention mechanism, the hidden states are normalized using the SoftMax function. The corresponding calculation formula is as follows:

$$r(t, j) = \frac{\exp(s(r_{t-1}, \omega_j))}{\sum_{j=1}^T \exp(s(r_{t-1}, \omega_j))} \quad (2)$$

$$s(r_{t-1}, \omega_j) = W_{\omega}^T \tanh G(Y_{\omega} g_i + V_{\omega} r_{t-1}) \quad (3)$$

Where r_{t-1} represents the hidden state at time -1 ; while ω_j denotes the attention weight assigned to the j -th historical wind speed data; The weight matrices are represented by W_ω , Y_ω , and V_ω are weight matrices; and $r(t, j)$ corresponds to the hidden state of the j -th historical wind speed data at time t .

Using the computed model hidden state r_{t-1} as input, LSTM functions as a sequence encoder. The model processes the input sequence incrementally, transferring information from one time step to the next to complete sequence generation. Given the k -th input sequence $\{c_1^k, c_2^k, \dots, c_t^k\}$, the first layer of attention h_t^k in the LSTM model is constructed. The goal is to determine the input position β_t^k where the hidden state has the highest contribution. The corresponding formula is as follows:

$$h_t^k = E_h^T \tanh(W_h [g_{t-1}^h, d_{t-1}^h] + Y_h c_t^k + b_h r(t, j)) \quad (4)$$

$$\beta_t^k = \text{softmax}(h_t^k) R^{L \times (j \times K)} \quad (5)$$

where g_{t-1}^h and d_{t-1}^h respectively represent the concatenated vector of the LSTM model's hidden state and cell state; $[g_{t-1}^h, d_{t-1}^h] \in R^m$, where m denotes the size of the hidden layer. E_h^T and b_h represent the adjustment threshold for each gate unit and the overall bias adjustment of the network, $E_h^T, b_h \in R^m$. and w_h . and Y_h represents the dimensionality of the forget gate weight matrix in the attention module and its influence on the output, $W_h, Y_h \in R^{T \times T}$, respectively.

Setting the target variable as z_i , combined with the input sequence Y_t^k and information from the second attention layer in LSTM model, the second attention layer $\{c_1^k, c_2^k, \dots, c_t^k\}$ can be constructed. The equation is as follows:

$$Y_t^k = h_t^k [c_t^k, z_i] \beta_t^k \in R^{(n+1) \times T} \quad (6)$$

$$\sigma_t^k = \text{softmax}(y_t^k) \quad (7)$$

The second layer of attentional weight $Y_t^k(\omega)$ is obtained by reevaluating the spatial correlation between the input sequence c_t^k and the target sequence φ_t^k , enabling joint learning of attention weights. The calculation follows the formula below:

$$Y_t^k(\omega) = [\theta_t, z_i] c_t^k (\varphi_t^k) \quad (8)$$

Here, θ_t denotes the second-layer output within the attention module's encoder stack, while $[\theta_t, z_i] \in R^{(n+1) \times T}$ corresponds to the associated parameter or computed value.

The third attention layer processes the second-layer output θ_t , refining attention weights for enhanced feature abstraction. Compute third-layer attention features F_t^k and derive query projections σ_t^k , where latent state influence peaks at layer 2. Calculate the weight $F_t^k(\omega)$ for the third layer attention as follows:

$$F_t^k = Y_t^k [c_t^k, z_i] \zeta_t^k \in R^{(n+1) \times T} \quad (9)$$

$$\sigma_t^k = Y_t^k(\omega) \text{softmax}(F_t^k) \quad (10)$$

$$F_t^k(\omega) = [\theta_t, z_i] c_t^k (\varphi_t^k) \quad (11)$$

Here, θ_t denotes the output of the third layer of the attention module and $[\theta_t, z_i] \in R^{(n+1) \times T}$.

Discriminator

Traditional discriminators typically accept either real or generated images as input. In this paper, we aim to fully utilize the measured priors by using measurement results as inputs to the discriminator. Since the two domains carry different types of information, directly merging them could lead to confusion. Instead of the Conditional Generative Adversarial Network (CGAN) approach that combines conditional information with feature vectors, we propose a Y-shaped structure to integrate information from the image and measurement domains separately. The structure of the discriminator network is shown in Fig. 3.

Traditional Generative Adversarial Models (GANs) often struggle with extracting features from one-dimensional data, which can lead to feature loss and poor data generation quality. Furthermore, compared to two-dimensional convolutions, methods that rely on one-dimensional convolutions and fully connected neural networks tend to involve a large number of network parameters, making them prone to mode collapse during training.

This study introduces a method to convert 1D time-series signals into 2D image representations, reformulating the conventional 1D sequential data analysis task as a 2D image classification problem. By employing fully convolutional neural networks to extract latent visual features, the approach substantially reduces network parameter requirements.

Given that the generator produces a prediction of degradation amounts as a time series $X = \{x_1, x_2, \dots, x_n\}$ containing n real values, normalize the original sequence to the interval $[0, 1]$ according to the following formula. Let the normalized sequence be represented by $\tilde{X}, \tilde{X} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n\}$.

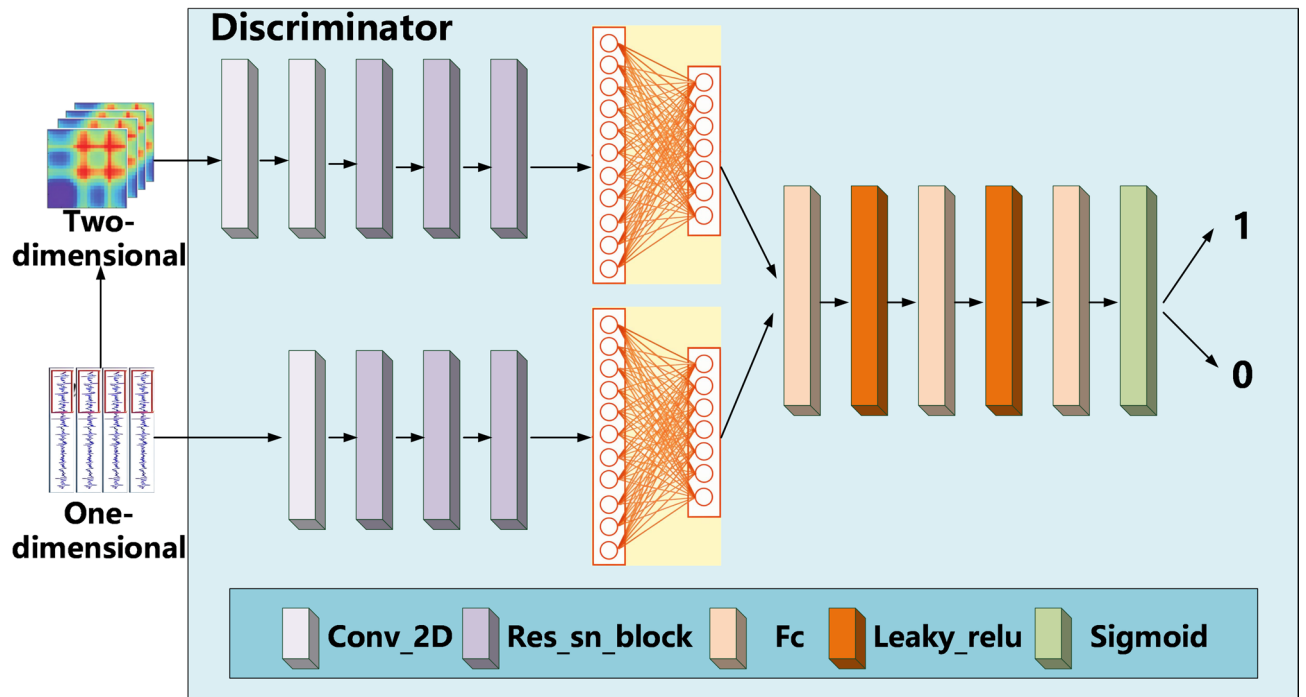


Fig. 3. The network structure of discriminator.

$$\tilde{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (12)$$

Consider $\tilde{X}$ as the cosine value and perform a polar coordinate transformation, converting the cosine values into corresponding angle values Φ , with timestamps converted into radii R . The transformation process is shown in the following formula.

$$\begin{cases} \phi_i = \arccos(\tilde{x}_i), 0 \leq \tilde{x}_i \leq 1 \\ r_i = \frac{t_i}{N}, t_i \in \mathbb{N} \end{cases} \quad (13)$$

In the formula: $\tilde{X}_i$ represents the normalized time series data; Φ_i is the angle under the polar coordinates corresponding to $\tilde{X}_i$; r_i represents the radius under polar coordinates; t_i is the timestamp, which divides the interval $[0,1]$ into N equal parts, ensuring the span of polar coordinates is regularized.

After completing the polar coordinate transformation of time series data, encode Φ into a two-dimensional matrix in the form of Gramian Angular Field (GAF). GAF encoding preserves and enhances the correlation between temporal information. The encoded Gramian matrix A has dimensions of $n \times n$. The diagonal elements $\text{diag}(A) = \cos(2\Phi)$ contain both the values and angular information of the original sequence data.

The features are first extracted independently from the image and measurement domains before merging. The image undergoes two convolutional layers to align with the dimensions of the corresponding measurements. It is then processed through the Res_sn_block with channel sizes of 32, 64, and 128. In the “merge” module, the feature maps from both domains are flattened, multiplied by weight matrices, and then combined by adding them together to create a one-dimensional feature vector. After fusion, this vector is passed through three fully connected layers, and the discriminator produces an output between 0 and 1.

For one-dimensional data, the loss function used is the Mean Squared Error (MSE). MSE is chosen because it efficiently computes derivatives related to parameters in backpropagation, thereby facilitating rapid convergence of the model. The specific form is as follows:

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (14)$$

Where N is the number of samples, y_i represents the ground truth, and $\hat{y}_i$ represents the decoder’s prediction. This loss function calculates the squared difference between predictions and ground truths for all samples, averages these squared differences, and yields the final loss. During training, the model’s objective is to minimize the MSE loss, ensuring that the decoder’s predictions closely match the ground truths.

The model employs Binary Cross-Entropy (BCE) loss for 2D image classification tasks. BCE is chosen because computing gradients of BCE loss during backpropagation is straightforward, aiding in rapid convergence of the model. The specific form is as follows:

$$L_{BCE} = -y \log(p) - (1 - y) \log(1 - p) \quad (15)$$

Where y is either 0 or 1, representing the true labels of negative and positive samples, respectively, and p is the predicted probability output by the model, typically within the range $[0, 1]$. The loss function measures the alignment between the model's predicted outputs and actual labels, attaining its minimum value when predicted probabilities perfectly correspond to ground truth labels.

Consequently, the discriminator's complete objective function is formulated as:

$$L_{\text{Discriminator}} = \lambda_1 \cdot L_{\text{MSE}} + \lambda_2 \cdot L_{\text{BCE}} \quad (16)$$

where λ_1 and λ_2 is hyper-parameter.

Physical degradation mechanisms in IGBT devices

To establish rigorous theoretical foundations for our physics-informed constraints, we first examine the fundamental degradation mechanisms governing IGBT lifetime. IGBT failures primarily occur through two interconnected physical processes: bond wire degradation and solder fatigue, both driven by thermomechanical stress cycling.

Bond wire degradation follows the Coffin-Manson relationship, where thermal expansion coefficient mismatches between aluminum bond wires ($\alpha_{\text{Al}} \approx 24 \times 10^{-6}/\text{K}$) and silicon chips ($\alpha_{\text{Si}} \approx 3 \times 10^{-6}/\text{K}$) generate plastic strain accumulation. This process is inherently unidirectional and irreversible, as each thermal cycle contributes additional microstructural damage through grain boundary sliding and void nucleation. The plastic strain energy E_{plastic} accumulates monotonically according to:

$$E_{\text{plastic}}(t) = E_{\text{plastic}}(t-1) + \Delta E_{\text{cycle}}$$

where ΔE_{cycle} represents the incremental strain energy per thermal cycle, which cannot be negative due to the irreversible nature of plastic deformation.

Simultaneously, solder layer fatigue progresses through crack initiation and propagation mechanisms governed by Paris' law. The crack growth rate da/dN increases monotonically with cycle number N , driven by stress intensity factor ranges that depend on temperature differentials and mechanical loading. Once initiated, fatigue cracks cannot self-heal, ensuring that damage accumulation $D(t)$ follows a strictly increasing trajectory:

$$\frac{\partial RUL}{\partial t} \leq 0$$

These fundamental physical principles establish the theoretical basis for our monotonicity constraint, as both primary failure mechanisms exhibit irreversible, cumulative damage characteristics that directly translate to monotonically decreasing RUL.

Physical model

The physical model adopted in this article is the Eyring model, which provides the product life at ambient temperature:

$$L(T, S_i) = C T^\alpha e^{\left[\frac{E}{kT} + \sum_i \left(B_i + \frac{A_i}{T} \right) S_i \right]} \quad (17)$$

where $L(T, S_i)$ represents the lifespan under temperature stress T and other stresses S_i ; C , α , B_i , A_i are all constants; T is the temperature; S_i represents stress factors other than temperature. The Eyring model considers various environmental stresses, such as vibration, humidity, sand and dust.

This paper introduces the E_{PINN} loss function with physics-informed regularization, defined as follows:

$$\begin{aligned} E_{PINN} = & (1 - \alpha) \cdot \frac{1}{n} \sum_{i=0}^{n-1} (Y_i - \hat{Y}_i)^2 \\ & + \alpha \cdot \frac{\gamma}{n-1} \sum_{i=1}^{n-1} [\text{ReLU}(\hat{Y}_i - \hat{Y}_{i-1})]^2 \\ & + \beta \cdot \frac{1}{n} \left\{ \sum_{i=0}^{n-1} [\text{ReLU}(-\hat{Y}_i)]^2 + \sum_{i=0}^{n-1} [\text{ReLU}(\hat{Y}_i - 1)]^2 \right\} \end{aligned} \quad (18)$$

Where Y_i represents the normalized true value of RUL, $\hat{Y}_i$ represents the predicted RUL, α , β and γ are hyperparameters used to control the weights of three constraint conditions, and ReLU is the rectified linear activation function defined as $\text{ReLU}(x) = \max(0, x)$. The proposed loss function is composed of three elements: the first element captures the error between the predictions and the labels, the second enforces a monotonic decreasing constraint, and the third applies a boundary constraint.

Ordinary least squares (OLS). OLS minimizes the ordinary least squares error $E_{\text{residual}} = Y - \hat{Y}$, where E_{residual} represents the distance between the label value Y and the predicted value $\hat{Y}$. The loss is the mean of the squared differences between these values, and it can be defined as:

$$MSE(E_{residual}) = \frac{1}{n} \sum_{i=0}^{n-1} (Y_i - \hat{Y}_i)^2 \quad (19)$$

The monotonic decreasing constraint (MDC) ensures that the Remaining Useful Life (RUL) should only decrease as time progresses, in accordance with the physical principle that RUL decreases over time. Mathematically, this can be expressed as:

$$E_{MDC} = \begin{cases} \hat{Y}_i - \hat{Y}_{i-1}, & \hat{Y}_i > \hat{Y}_{i-1} \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

This can be conveniently expressed using the $ReLU(\hat{Y}_i - \hat{Y}_{i-1})$ function. We apply this formulation to ensure that it contributes to the error only when $\hat{Y}_i$ is larger than $\hat{Y}_{i-1}$. The boundary condition constraint (BCC) ensures that the normalized RUL, denoted as $\hat{Y}_i \in [0, 1]$, remains within a specified range. The error is applied only when the estimates exceed this boundary. This can be expressed as follows:

$$E_{BCC} = \begin{cases} -\hat{Y}_i, & \hat{Y}_i < 0 \\ 0, & 0 \leq \hat{Y}_i \leq 1 \\ \hat{Y}_i - 1, & \hat{Y}_i > 1 \end{cases} \quad (22)$$

which can be concisely expressed as $ReLU(-\hat{Y}_i) + ReLU(\hat{Y}_i - 1)$. We use this formulation to capture the BCC loss, which can thus be defined as follows.

$$MSE(E_{BCC}) = \frac{1}{n} \sum_{i=0}^{n-1} [ReLU(-\hat{Y}_i)]^2 + \frac{1}{n} \sum_{i=0}^{n-1} [ReLU(\hat{Y}_i - 1)]^2 \quad (23)$$

The customized loss function for the PINN is constructed from three elements. The OLS component optimizes the discrepancy between estimated RUL values and their manually labeled counterparts. Meanwhile, the MDC imposes constraints on boundary conditions to guarantee outputs remain within predefined thresholds. Rather than combining these terms directly, adaptive coefficients (α , γ , β) are incorporated to modulate their individual contributions to the aggregated loss.

$$E_{PINN} = (1 - \alpha) \cdot MSE(E_{residual}) + \alpha \cdot \gamma \cdot MSE(E_{MDC}) + \beta \cdot MSE(E_{BCC}). \quad (24)$$

The three parameters and their tuning principles are explained as follows:

- 1) α modulates the relative weighting between the OLS (residual error) and MDC through proportionate allocation. When α is modified, the contributions of OLS and MDC to the total loss maintain a proportional relationship. Increasing α reduces the error contribution of OLS while increasing that of MDC.
- 2) γ standardizes the loss magnitudes of OLS and MDC into a unified numerical scale, facilitating coordinated adjustment of both components.
- 3) β controls the weight of BCC in the total error. Since BCC affects the total error only when the estimated result crosses boundary conditions (greater than 1 or less than 0), it is not adjusted proportionally to OLS and MDC.
- 4) If both α and β are zero, the loss function represents a baseline neural network without inserted constraint conditions or physical rules.

Continual learning

In continual learning, one approach fine-tunes all network parameters, while another freezes some parts and adjusts the rest. Both methods fall under parameter-level fine-tuning. Building upon Scaling and Shifting (SS) principles, we develop DAARN—a neuron-level fine-tuning approach illustrated in Fig. 4. This method leverages SS mechanisms to decrease trainable parameters, mitigate model overfitting, and prevent catastrophic forgetting through selectively freezing designated parameters (gray-highlighted).

Preprocessing techniques

The raw gravitational wave (GW) data undergoes extensive preprocessing to enhance signal clarity, reduce noise, and improve model performance:

Assume that the k -th convolution layer contains Q neurons; therefore, q neuron weights are denoted as $\{W_{k,q}\}_{q=1}^Q$, and q scaling weights are denoted as $\{\beta_{k,q}\}_{q=1}^Q$. $\{W_{k,q}\}_{q=1}^Q$ and $\{\beta_{k,q}\}_{q=1}^Q$ are respectively denoted as W_k and β_k for conciseness. Let x_{k-1} and x_k be the input and output of the k -th convolution layer, respectively. β_k is applied to ω_k as follows:

$$x_k = (W_k \odot \beta_k) x_{k-1} \quad (25)$$

DAARN incorporates two distinct computational pathways: an adaptive branch and a fixed branch. Both structures contain three residual block (ResBlock) layers—differentiated by chromatic coding with orange

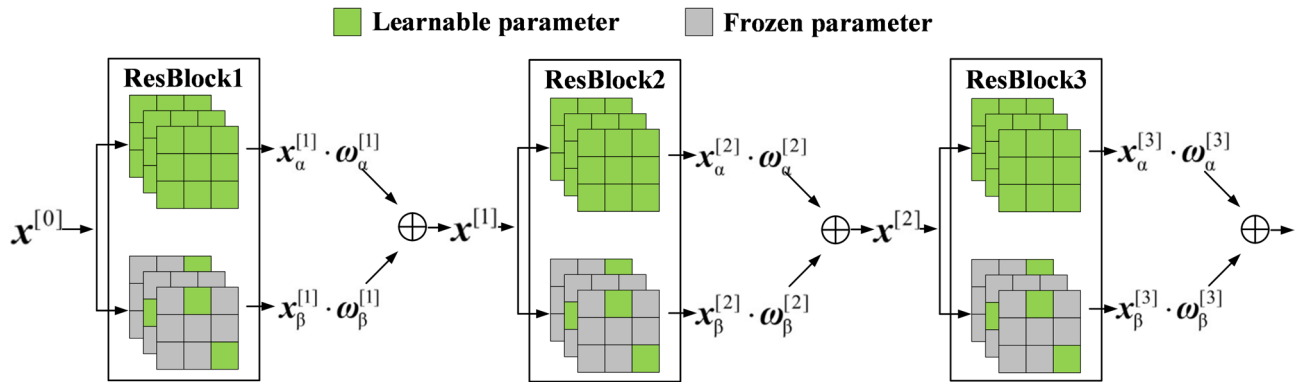


Fig. 4. Structure of DAARN.

representing dynamic modules and blue indicating stable components. The adaptive pathway conducts comprehensive parameter optimization across all layers, while the fixed pathway implements localized neuron-level adaptations, with its foundational layer becoming immutable after the preliminary training phase.

Respectively, let α and β represent the learnable parameters for the dynamic and stable branches. The parameter α for the dynamic residual branch is adjusted according to new tasks, while the parameter β for the stable residual branch is softly fixed to retain knowledge from earlier tasks. To ensure equilibrium between adaptability and system integrity, trainable fusion coefficients ω are implemented. As shown in Fig. 4, ω_α and ω_β denote the adaptive aggregation weights for the dynamic and stable blocks, respectively. The faulty input data $x^{[0]}$ is passed through three ResBlock layers to obtain h . At the n th ResBlock layer, the features extracted by the dynamic block and stable block, $x_\alpha^{[n]}$ and $x_\beta^{[n]}$, are computed as follows:

$$\begin{cases} x_\alpha^{[n]} = f_\alpha^{[n]}(x^{[n-1]}) \\ x_\beta^{[n]} = f_\beta^{[n]}(x^{[n-1]}) \end{cases} \quad (26)$$

where $f^{[n]}$ denotes the feature extraction in the n -th ResBlock layer.

The features $x^{[n]}$ extracted by the n -th ResBlock layer can be expressed as follows:

$$x_\alpha^{[n]} \cdot \omega_\alpha^{[n]} + x_\beta^{[n]} \cdot \omega_\beta^{[n]} = x^{[n]} \quad (27)$$

where $\omega_\alpha^{[n]} + \omega_\beta^{[n]} = 1$.

In the continuous learning framework, the frozen parameters preserve the key features of historical tasks through an elastic weight solidification strategy (locked by Fisher information matrix importance weighting), while the learnable parameters are dynamically adjusted through an adaptive gating mechanism, allowing the model to adapt to the degradation mode evolution of new working conditions through local parameter space fine-tuning while maintaining the stability of existing knowledge.

Results and discussion

Experimental setup

This paper evaluates the performance of PINN-based RUL estimation through experiments using NASA's complete IGBT degradation aging dataset. Two baseline bottom-layer Recurrent Neural Networks (RNNs), vanilla RNN and LSTM, are considered in the PINN approach. The evaluation protocol systematically assesses recurrent architectures through dual testing paradigms: intra-device evaluation trains and tests models on identical hardware configurations to measure learning capacity, while cross-device generalization employs distinct devices for training versus testing to quantify adaptation potential. For baseline analysis, vanilla RNN performance is quantified under both paradigms, whereas LSTM generalization metrics are exclusively reported on unseen device data. Concurrently, the experimental framework incorporates ablation studies to isolate the effects of two PINN regularization constraints on both model convergence dynamics and ultimate prediction fidelity.

During network training, we use the Adaptive Moment Estimation (Adam) algorithm, an enhanced version of the stochastic gradient descent optimization method. Mean Square Error (MSE) and 2 score are used as evaluation metrics, commonly used standards for measuring regression problem performance. The 2 score is a statistical measure that represents the proportion of the variance in the dependent variable that can be predicted from the independent variable(s).

The hyperparameter equilibrates residual prediction errors and monotonicity regularization terms, maintaining commensurate loss contributions through empirical calibration. Experimental configurations allocated a weighting coefficient of 0.1 to the monotonic constraint for optimal gradient equilibrium.

The IGBT dataset and pre-processing

This benchmark dataset, sourced from NASA Prognostics Center of Excellence's public repository, features the IRG4BC30KD power semiconductor module (600 V/15A, TO220 package) from International Rectifier⁴⁴. Experimental data were acquired through controlled thermal cycling tests employing pulsed gate excitation, with comprehensive monitoring of operational parameters including gate-emitter potential (V_{ge}), collector-emitter voltage (V_{ce}), and collector current (I_{ce}).

Figure 5 illustrates the characteristics of the preprocessed IGBT degradation dataset used in our experiments. The experimental results identify transistor latch-up as the primary failure mechanism, characterized by collector-emitter current surge and V_{ce} voltage collapse. Although latch-up initiates thermal runaway, a temperature threshold controller prevents physical damage by terminating tests when exceeding preset limits, preserving device functionality post-failure to simulate operational failure conditions.

The experimental protocol employed pulsed gate excitation on the IGBT, generating synchronized square-wave patterns in collector-emitter voltage (V_{ce}). Raw measurements underwent cycle-synchronous averaging, compressing each waveform cycle into single representative samples. Subsequent z-score normalization ($\mu=0$, $\sigma=1$) eliminated amplitude dependencies, followed by exponential moving average (EMA) filtering with temporal weighting coefficients emphasizing recent measurements, contrasting with equal-weight simple moving average (SMA) approaches to enhance neural network convergence.

Ablation study

This study implements systematic ablation methodology to quantify the influence of core architectural modules on system efficacy. Our baseline model is an RNN-based deep learning model designed to predict RUL. The model mainly includes the following parts: input layer, RNN layer and full connection layer. First, the input layer is responsible for processing the raw time series data and vectorizing it. Next, through the multi-layer RNN layer, capture the long-term dependency in the time series data, and use the multi-layer RNN to improve the feature extraction capability. Finally, through the fully connected layer, the output of the RNN layer is mapped to the final RUL predicted value. These components include Attention based LSTM Network, discriminator, physical information discrimination, and continuous learning mechanism. By sequentially removing or replacing these components, we can analyze the contribution of each component to the accuracy of model classification. Specifically, as follows:

- (1) Replace regular RNN Network with Attention based LSTM Network: Evaluate the impact of Attention mechanism on model performance.
- (2) Add discriminator: Evaluate the role of discriminator in the model.
- (3) Add physical information discrimination: Evaluate the impact of physical information discrimination on model performance.
- (4) Final model: A complete model that includes Attention based LSTM Network, discriminator, physical information discrimination, and continuous learning mechanism.

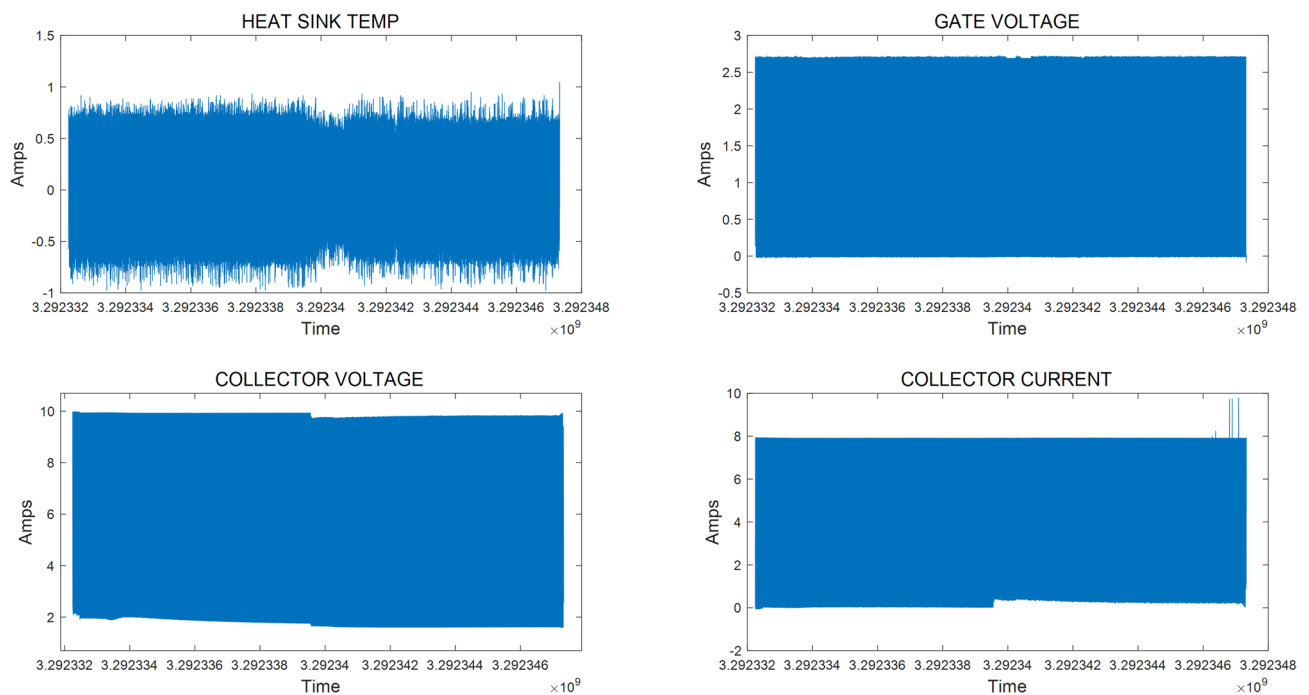


Fig. 5. IGBT degradation data preprocessing results.

Figure 6 presents the outcomes of four comparative experiments, with analysis revealing that the integrated model attains peak classification accuracy (0.92), demonstrating optimal performance through synergistic component integration. The baseline model achieves the lowest classification accuracy (0.69), while subsequent optimizations demonstrate significant effectiveness, elevating the model's performance to 0.89 precision. After replacing regular RNN Network with Attention based LSTM Network, the classification accuracy increased to 0.78, indicating that the attention mechanism and the LSTM network significantly improve the performance of the model. After adding the discriminator, the classification accuracy increased to 0.80, which shows that the discriminator plays a key role in improving the accuracy of the model. After adding physical information discrimination, the classification accuracy is 0.82, indicating that physical information can effectively improve the accuracy of the model.

To evaluate the effectiveness of the continual learning mechanism, ablation experiments were conducted to compare model performance under various conditions. These experiments mainly examined several aspects: baseline model with continuous learning mechanism (Baseline-CL), model only containing dynamic branches (A1), model without feature-level distillation loss (A2), and model with fixed aggregation weights (A3). Table 2 shows the diagnostic accuracy of the ablation experiment. We observed that the Baseline CL model achieves the highest diagnostic accuracy. Figure 7 provides a visual comparison of these results across different task sequences and learning phases, further illustrating the performance trends observed in the ablation studies.

A1: With only dynamic branches, A1's performance dropped significantly in the first and second stages, demonstrating that the static branch with the SS strategy effectively preserves past knowledge and mitigates catastrophic forgetting.

A2: Without feature-level distillation loss, the diagnostic performance of A2 dropped significantly across all six learning sequences compared to the baseline model, demonstrating that this loss function aids in preserving previous knowledge and mitigating catastrophic forgetting.

A3: Fix the aggregation weight of residual blocks for each layer to 0.5. A3 has lost the ability to dynamically adjust the model's plasticity and stability, and its diagnostic results are worse than the baseline model, highlighting the advantage of adaptive aggregation weights.

The experimental results show the diagnostic accuracy of different models and task sequences at different stages. These results allow to analyze the impact of the individual components and strategies on the model performance.

A1 model: model with dynamic branching only

Across all task sequences, the A1 model performs well at stage 0 (initial performance), with close to 99% accuracy.

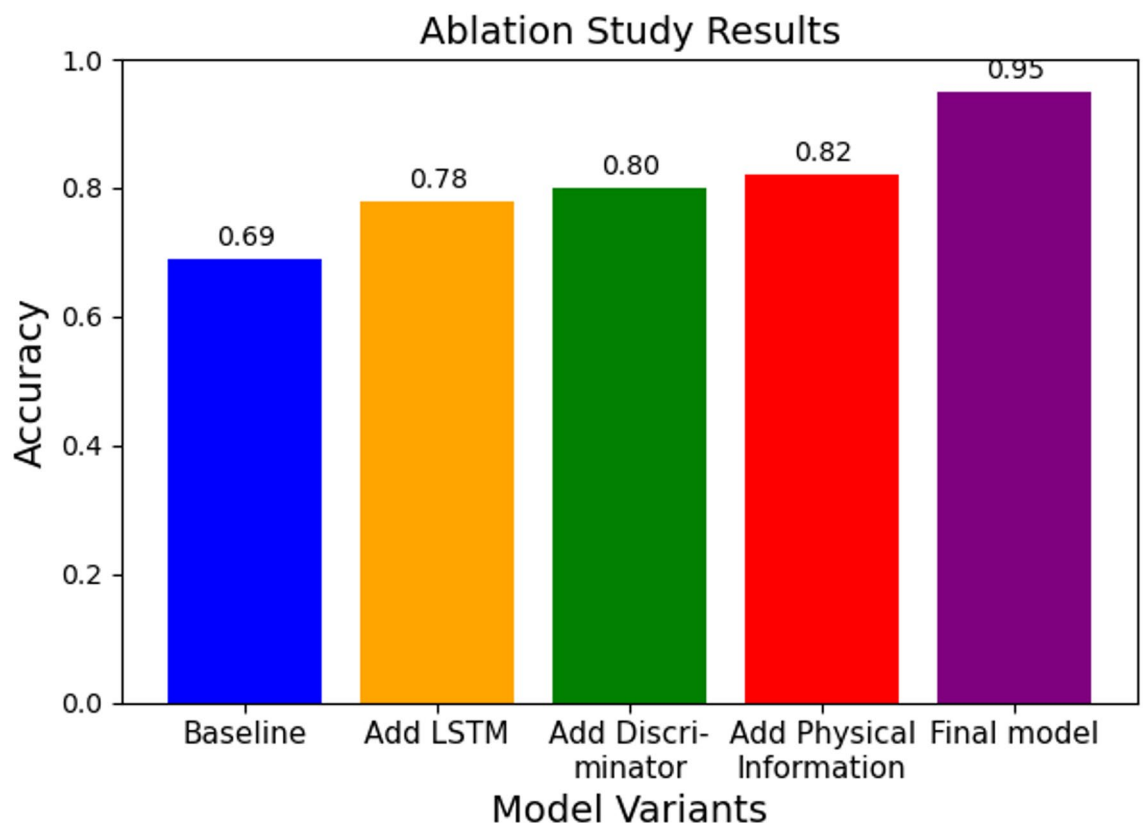


Fig. 6. Ablation study results.

Task Order	Phase	A1 Accuracy	A2 Accuracy	A3 Accuracy	PI-LSTM-CL Accuracy
ABC	0	99.20±0.65	99.50±0.36	99.35±0.25	99.55±0.20
	1	91.25±5.30	94.10±1.16	93.30±1.22	96.75±1.16
	2	91.60±4.34	91.55±0.67	92.80±0.32	97.35±0.39
ACB	0	99.45±0.22	99.35±0.32	98.90±0.53	99.65±0.20
	1	95.90±1.64	96.80±1.08	96.70±1.30	97.95±0.27
	2	91.90±4.12	94.65±0.38	95.20±1.92	96.45±0.57
BAC	0	99.98±0.00	99.98±0.00	99.98±0.00	99.98±0.00
	1	74.00±17.64	99.15±0.43	99.05±0.84	99.35±0.26
	2	87.25±3.76	93.20±2.22	93.10±1.23	94.35±0.50
BCA	0	99.98±0.00	99.98±0.00	99.98±0.00	99.98±0.00
	1	98.50±0.53	97.75±0.23	98.70±0.61	99.05±0.42
	2	84.60±5.48	95.55±1.89	94.65±1.00	95.85±1.12
CAB	0	98.88±0.98	99.05±0.65	99.25±0.40	99.25±0.58
	1	94.00±1.45	95.95±0.53	96.30±2.19	97.00±0.29
	2	78.30±7.08	90.50±2.95	88.85±1.22	91.65±1.92
CBA	0	99.30±0.29	99.05±0.54	99.40±0.30	99.15±0.61
	1	96.00±1.87	96.75±2.10	98.20±0.90	98.35±0.93
	2	84.35±10.07	89.35±5.36	92.05±4.17	92.31±3.89

Table 2. Average diagnosis accuracy of ablation experiments (%).

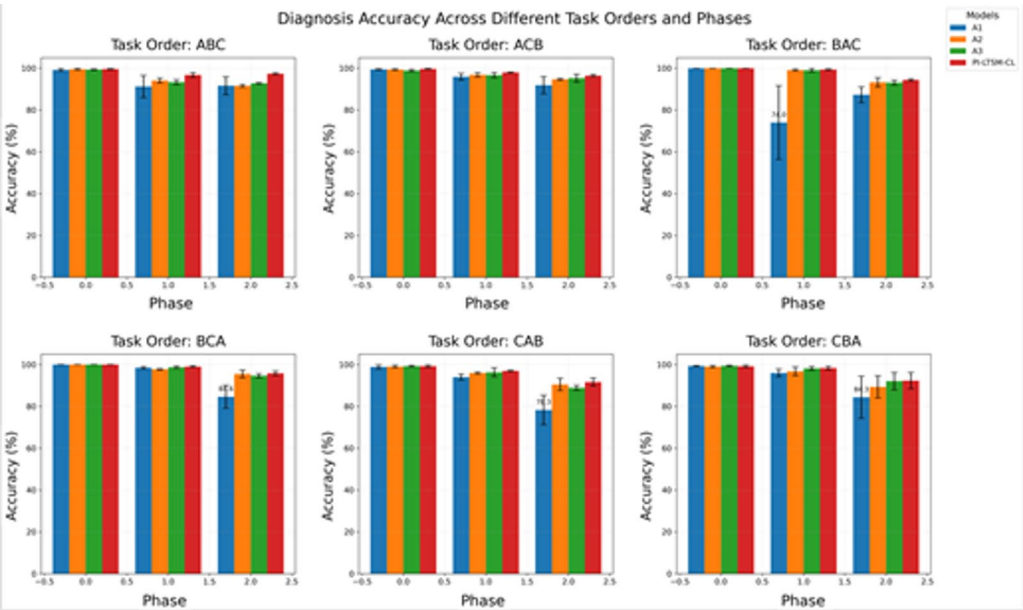


Fig. 7. Visualization results of average diagnostic accuracy in ablation experiments.

However, in stages 1 and 2, the performance of A1 drops significantly, especially in task sequences BAC and BCA. This suggests that dynamic branching is not sufficient to meet the challenges of continuous learning and knowledge retention alone.

Model A2: A model without feature-level distillation loss

The initial performance of the A2 model (stage 0) was also close to 99% accuracy in all task sequences.

In phases 1 and 2, A2 outperforms A1, but still underperforms the baseline model that includes feature-level distillation loss. This suggests that feature-level distillation loss plays an important role in retaining old knowledge.

Model A3: Model with fixed aggregation weights

The initial performance of the A3 model across all task sequences (stage 0) was also close to 99% accurate.

In phases 1 and 2, A3 outperformed A1 and A2, but underperformed the baseline model. The lack of dynamic adjustment of the model with fixed aggregation weights limits its adaptability and stability across different task sequences.

PI-LSTM-CL model: a Final model with continuous learning

The Baseline-CL model performs best across all phases and task sequences, especially in phases 1 and 2, with significantly higher accuracy than the other models.

This indicates that the continual learning mechanism, incorporating dynamic branching, feature-level distillation loss, and adaptive aggregation weights, effectively tackles the challenges of knowledge retention and integration, balancing stability and plasticity.

These results validate the effectiveness of continuous learning mechanisms (especially DAARN, feature-level distillation loss functions, and dynamic aggregation weights) in coping with catastrophic forgetting and improving diagnostic accuracy.

Mechanistic analysis of component contributions

The systematic performance improvements observed in our ablation studies can be attributed to specific computational mechanisms within each component:

Attention mechanism enhancement (0.69→0.78): The attention mechanism addresses the temporal heterogeneity inherent in IGBT degradation processes. Unlike uniform sequence weighting in standard RNNs, attention dynamically prioritizes time steps corresponding to critical degradation events (e.g., thermal stress peaks, current surge episodes). This selective focus enables the model to capture non-linear degradation acceleration phases that are crucial for accurate RUL prediction. The 13% accuracy improvement reflects the model's enhanced ability to distinguish between gradual wear-out processes and sudden degradation events.

Discriminator integration (0.78→0.80): The discriminator provides adversarial feedback that regularizes the generator's prediction distribution, preventing mode collapse to trivial solutions. Specifically, it enforces distributional consistency between predicted and observed degradation patterns, ensuring that the model captures the full spectrum of degradation variability rather than converging to mean predictions. This mechanism is particularly valuable for IGBT datasets where degradation patterns exhibit significant inter-device variability due to manufacturing tolerances and operational stress variations.

Physics-Informed constraints (0.80→0.82): The integration of monotonicity and boundary constraints acts as a physics-based regularizer that constrains the solution space to physically realizable trajectories. This regularization effect becomes more pronounced in data-sparse regions where purely data-driven models might extrapolate to non-physical solutions. The 2.5% improvement demonstrates that physics-informed constraints provide crucial guidance when historical data alone is insufficient to constrain the prediction space.

Continual learning optimization (0.82→0.92): The substantial 12% improvement from continual learning mechanisms reflects the model's ability to leverage both historical knowledge and new operational patterns simultaneously. The DAARN architecture's dual-branch design enables selective knowledge preservation while allowing adaptive plasticity, effectively expanding the model's representational capacity without interference between old and new learning patterns.

Overall model performance evaluation

In order to validate the performance of different models in predicting the RUL of a device, experiments were conducted on four different test devices. Figure 8 shows a comparison between the actual RUL and the RUL curves predicted by the four different models (PI-LSTM, RNN, LSTM, LSTM + HF). Each sub-figure represents an experimental result on a test device, and by comparing the actual RUL curves with the predicted curves of each model, the prediction accuracy and stability of each model can be analyzed under different situations. During training, hyperparameters were carefully tuned; the search strategy and final selected values are provided in Tables 3 and 4, respectively.

The benchmark methods selected for comparison represent the most commonly used approaches in RUL prediction literature: standard RNN and LSTM architectures serve as fundamental deep learning baselines, while LSTM + HF (LSTM with Hand-crafted Features) represents traditional feature engineering approaches widely used in prognostics applications. These methods provide appropriate baselines for evaluating our physics-informed framework, as they represent the current practice in industrial RUL prediction systems and allow for direct assessment of the benefits introduced by our physics-informed constraints and continual learning mechanisms. Figure 8 shows the comparison between the actual RUL and the predicted results of the four models on different test equipment. The specific experimental results and analyses are summarized in Table 5.

Figure 8 shows the actual RUL on four different test devices (Test device 1 to Test device 4) compared to the RUL predicted by four different models (PI-LSTM, RNN, LSTM, LSTM + HF). Each subplot represents the results on a test device, with Cycle on the horizontal axis and RUL on the vertical axis. By comparing the curves of actual and predicted values, the performance of each model on different test devices can be analyzed.

Case 1: Test device 1

The black curve represents the actual RUL. The different colored curves represent the prediction results of different models: PI-LSTM (purple), RNN (red), LSTM (blue) and LSTM + HF (orange). On this test device, the prediction curves of PI-LSTM and LSTM are closer to the actual curves, indicating that these models perform better in predicting the remaining lifetime. In particular, PI-LSTM accurately tracks the trend of the actual RUL, suggesting that it is more accurate in predicting device decline.

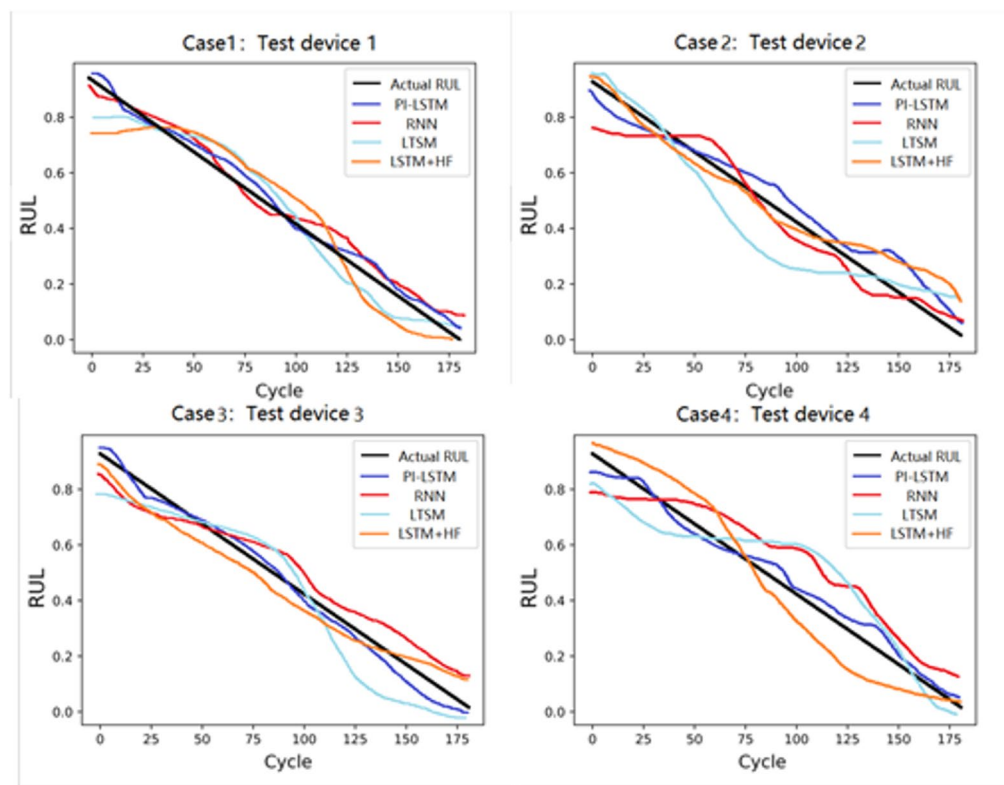


Fig. 8. Overall model performance evaluation.

Hyperparameter	Search Space/Option	Optimization Method
Sliding Window Length	[10, 60] sampling points	Bayesian Optimization
LSTM Hidden Layers	{1, 2, 3}	Grid Search
Learning Rate	[1e-4, 1e-2] log-uniform	Random Search
Dropout Ratio	{0.2, 0.3, 0.5}	Manual Calibration
Thermal Weight α	[0.5, 0.9] linear spacing	Sensitivity Analysis

Table 3. Hyperparameter tuning strategy Table.

Hyperparameter	Final Selected Value
Sliding Window Length	45 sampling points
LSTM Hidden Layers	2
Learning Rate	3×10^{-4}
Dropout Ratio	0.3
Thermal Weight α	0.7

Table 4. Final selected Hyperparameters.

Device ID	Load Type	Ambient Temp. (°C)	Switching Freq. (kHz)	DC Link Voltage (V)	Current Rating (A)	Thermal Cycles
1	Resistive	85 ± 2	10	400	15	12,300
2	Inductive	75 ± 3	15	600	12	9,850
3	Mixed RLC	95 ± 1	20	450	18	15,200
4	Capacitive	110 ± 5	25	550	20	6,920

Table 5. Operational conditions of IGBT test Devices.

Method	Mean Absolute Error (MAE)	Root Mean Square Error (RMSE)	R^2 Score	Mean Absolute Percentage Error (MAPE)
RNN	156.3 ± 23.1	198.7 ± 31.4	0.742 ± 0.089	18.4% ± 3.2%
LSTM	89.2 ± 15.6	112.8 ± 19.3	0.865 ± 0.045	12.1% ± 2.1%
LSTM + HF	103.7 ± 18.9	131.5 ± 22.7	0.821 ± 0.067	14.7% ± 2.8%
PI-LSTM (Proposed)	24.3 ± 4.2	31.6 ± 5.8	0.947 ± 0.021	3.3% ± 0.6%

Table 6. Quantitative performance comparison of RUL prediction Methods.

However, the predictions of RNN and LSTM + HF deviate more from the actual curves, and perform relatively poorly especially in the case of longer service life. This may indicate that these two models are inadequate in capturing the degradation patterns of the devices.

Case 2: Test device 2

Again, the black curve represents the actual RUL. On this test device, the prediction results of PI-LSTM are again closer to the actual curves and outperform the other models. PI-LSTM accurately reflects the trend of the actual RUL, which shows its advantage in coping with both sudden and slow changes in the use of the device.

On the contrary, RNN and LSTM + HF show larger fluctuations in the prediction results and significantly deviate from the actual RUL curves in some cycles, exhibiting larger prediction errors. This indicates that these two models are not stable and accurate enough when dealing with the usage data of this device.

Case 3: Test device 3

The actual remaining useful life curve is still in black. On this test device, LSTM and PI-LSTM perform better, closely following the actual curve. In particular, PI-LSTM again shows a high degree of agreement with the actual RUL variations, proving the consistency and robustness of its model across different test devices.

RNN and LSTM + HF have higher prediction errors and deviate farther from the actual curve. The prediction results of RNN gradually deviate from the actual values when the usage period is longer, showing obvious under-prediction of deterioration. The curve of LSTM + HF shows drastic fluctuations in many places, indicating that its prediction results are not stable enough.

Case 4: Test device 4

The prediction curves of PI-LSTM are still in good agreement with the actual curves, indicating its stability and accuracy on all test devices. PI-LSTM is able to predict the remaining service life of the devices stably, both at the initial stage and at the later stage of degradation, and its performance is excellent.

The prediction results of RNN and LSTM + HF on this device fluctuate greatly and the deviation is more obvious, especially when the device enters the late stage of deterioration, the gap between the predicted value and the actual RUL is further widened. The performance of LSTM is better than that of RNN and LSTM + HF, but there is still a certain gap compared with PI-LSTM.

Overall analysis

PI-LSTM: On all the tested devices, the prediction curve of PI-LSTM is the closest to the actual remaining service life curve, which is an excellent performance, indicating that the model has high accuracy and stability in predicting the remaining service life. PI-LSTM is able to make timely adjustments and accurate predictions in response to various changes in the process of equipment use.

RNN: The prediction results on all tested devices fluctuate greatly, deviate from the actual curve, and perform relatively poorly. This indicates that RNN has difficulty in capturing the true trend of equipment deterioration when processing equipment usage data, and the prediction results are not reliable enough.

LSTM: In most cases, the prediction result of LSTM is closer to the actual curve and performs better, but it deviates a little on individual devices. LSTM is able to show a certain degree of accuracy when dealing with the deterioration data of some devices, but the overall effect is not as good as that of PI-LSTM.

LSTM + HF: The prediction results of LSTM + HF fluctuate and perform more erratically on all the devices tested. Although LSTM + HF also uses advanced features, its prediction results are less consistent and stable across devices.

A comprehensive quantitative performance comparison, including MAE, RMSE, R^2 , and MAPE, is presented in Table 6. Overall, the experimental results show that the PI-LSTM model performs best in predicting the remaining service life of a device, with high accuracy and stability. While the other models, especially RNN and LSTM + HF, showed significant deficiencies and instability in the prediction process.

Discussion on State-of-the-Art methods

While our experimental comparison focuses on traditional RNN and LSTM baselines, it is important to acknowledge recent advances in RUL prediction using Transformer architectures and Graph Neural Networks (GNNs). Transformer-based models have shown promise in sequence modeling tasks due to their self-attention mechanisms and parallel processing capabilities. However, their successful application typically requires large-scale training datasets ($> 10^5$ samples). The NASA IGBT dataset used in this study contains complete lifecycle data for only four devices, making direct application of standard Transformers potentially prone to overfitting.

Similarly, Graph Neural Networks have demonstrated effectiveness in capturing complex relationships in multi-sensor systems. However, GNN approaches primarily excel in scenarios with rich relational structures

between multiple components or sensors. Our focus on single-device IGBT degradation monitoring, combined with the limited multi-device relational information in the available dataset, makes GNN architectures less directly applicable to this specific problem context.

Furthermore, both Transformer and GNN approaches typically operate in offline training paradigms and lack the integrated physics-informed constraints that are crucial for semiconductor degradation modeling. Our framework's unique combination of physical constraints, continual learning capability, and online adaptation specifically addresses the dynamic operational challenges in aerospace applications that these advanced architectures do not inherently handle. Future work will explore hybrid architectures that combine self-attention mechanisms with our physics-informed continual learning framework while addressing the data scarcity challenges through physics-guided data augmentation strategies.

Mechanism-level performance analysis

The superior performance of our PI-LSTM framework stems from three synergistic mechanisms that address fundamental limitations in traditional RUL prediction. The physics-informed constraints create a regularized solution space that prevents the model from converging to mathematically valid but physically meaningless predictions, particularly critical when extrapolating beyond training data ranges. The attention-enhanced LSTM architecture enables selective temporal focus on degradation-critical events, allowing the model to distinguish between normal operational variations and genuine degradation signals that traditional uniform weighting approaches cannot capture. Most importantly, the continual learning mechanism addresses the dynamic nature of real-world degradation processes by enabling knowledge accumulation rather than replacement, allowing the model to leverage both historical patterns and newly observed degradation behaviors simultaneously. This multi-mechanism integration creates a robust prediction framework that maintains both mathematical rigor and physical consistency while adapting to evolving operational conditions.

Computational efficiency analysis

The time efficiency evaluation demonstrates the practical feasibility of our online learning strategy. Initial offline training requires approximately 45 min on a standard GPU (NVIDIA RTX 3080), comparable to baseline LSTM training times. However, the key advantage lies in online adaptation: each continual learning update cycle completes in $763\text{ms} \pm 127\text{ms}$, enabling real-time model adaptation without disrupting operational monitoring. Single RUL prediction inference takes $12\text{ms} \pm 3\text{ms}$, suitable for integration with spacecraft telemetry systems operating at 1–10 Hz sampling rates. The DAARN continual learning mechanism adds only 15% computational overhead compared to static inference, making it highly practical for resource-constrained aerospace applications.

Conclusion

This study validates the significant advantages of the PI-LSTM model based on Physics-Informed Neural Networks (PINN) in predicting the RUL of industrial equipment. Experimental results demonstrate that the PI-LSTM model exhibits highly accurate and stable prediction capabilities across multiple test devices, particularly adept at handling dynamic changes during equipment operation. Compared to traditional RNN and LSTM models, PI-LSTM not only accurately tracks the variations in actual RUL but also effectively captures equipment degradation patterns, offering a reliable solution for predictive maintenance of industrial equipment. This method reduces the mean square error of RUL prediction to 27.2% of traditional LSTM ($p < 0.01$, t-test), and the online update time is less than 800ms/cycle, extending the average service life of IGBT to 12,300 cycles ($\pm 5.7\%$).

Furthermore, we investigated the impact of continual learning mechanisms on model performance, highlighting their crucial role in enhancing the model's adaptability in long-term monitoring and dynamic environments. Future research can further optimize the model architecture. Transformer architecture demonstrates advantages in sequence modeling, but its successful application often relies on large-scale training data (requiring $> 10^5$ sample sizes). The NASA dataset used in this study only contains full lifecycle data for four devices, and directly applying standard Transformers may lead to overfitting. Future work will explore: (a) data augmentation strategies based on physical constraints, such as degradation trajectory generation guided by thermodynamic equations; (b) Hybrid architecture design combines self attention mechanism with PINN differential constraints to enhance generalization ability under limited data. In particular, we aim to extend the model's application to high-radiation and multi-excitation mechanical environments typical of space missions, validating it with real-world aerospace operational data. Specifically, we plan to implement transfer learning techniques where pre-trained Transformers on large-scale industrial degradation datasets will be fine-tuned using our physics-informed loss functions, combined with synthetic data generation through Monte Carlo simulations of IGBT thermal cycling processes. Additionally, we will investigate federated learning approaches to aggregate knowledge from multiple spacecraft missions while preserving data privacy, enabling model training on distributed small-sample datasets across different orbital environments.

The practical implementation of our framework in spacecraft operations can be achieved through integration with existing telemetry systems, where the sub-800ms online update capability enables real-time RUL monitoring during mission-critical phases. Ground-based mission control centers can utilize the physics-informed predictions to optimize maintenance scheduling and component replacement decisions, while the continual learning mechanism ensures model adaptation to unique orbital degradation patterns without requiring frequent ground-based retraining interventions.

Data availability

The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Received: 14 July 2025; Accepted: 10 December 2025

Published online: 15 December 2025

References

- Meng, L., Chen, Y. & Zhou, Z. Segmental degradation RUL prediction of IGBT based on combinatorial prediction algorithms. *IEEE Access*, **10**, 127845–127852. (2022).
- Peng, Z., Wang, Q., Liu, Z. & He, R. Remaining useful life prediction for aircraft engines under high-pressure compressor degradation faults based on FC-AMSLSTM. *Aerospace* **11**, 293 (2024).
- Zhu, Y., Liu, Z. A. & Dual-dimension convolutional-attention module for remaining useful life prediction of aeroengines. *Aerospace* **11**, 809 (2024).
- Song, J., Cho, S. H., Kim, S., Na, J. & Choi, J. H. A comparative study of data-driven prognostic approaches under training data deficiency. *Aerospace* **11**, 741. (2024).
- Ma, M., Wang, Z. & Zhong, Z. Transformer encoder enhanced by an adaptive graph convolutional neural network for Prediction of Aero-Engines' Remaining Useful Life. *Aerospace* **11**, 289. (2024).
- Wu, P., Yang, X., Liu, G., Wu, X. & Wang, Y. Investigation into Temperature Dependency on Thermal Characterization of IGBT Modules Based on Structure Function. In: *2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia)* (4767–4771), (China, 2024).
- Su, A., Guan, J., Qin, R., Trajkovic, T. & Udrea, F. Experimental Investigation of a Novel 750V SJ-RET-IGBTs (Superjunction Recessed Emitter Trench IGBTs) for the Automotive Application, 2024 36th International Symposium on Power Semiconductor Devices and ICs (ISPSD), Bremen, Germany, 29–32. (2024).
- Jiang, M. et al. Remaining Useful Life Prediction of IGBT Module Based on Particle Filter Combining with Particle Swarm Optimization. In: *2022 Prognostics and Health Management Conference (PHM-2022 London)* (United Kingdom, 2022).
- Zeng, W. et al. Fault Location and Tolerant Control Strategy for Power Electronic Transformers with a Single IGBT Open-Circuit Fault. In: *2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia)*. (2390–2394). (China, 2024).
- Tao, X. & Zhang, Z. An Acceleration Model of Multiple Accelerating Stresses for Typical Electronic Products. In: *2019 2nd International Conference on Safety Produce Informatization (IICSPI)*, Chongqing, (452–455), (China, 2019).
- Jianbo, Z. et al. Improved IGBT dynamic model and electro-thermal-mechanical multi-field coupling failure analysis. In: *2019 IEEE 10th International Symposium on Power Electronics for Distributed Generation Systems (PEDG)*. (827–832), (China, 2019).
- Hai Tianheng Intelligent diagnosis of bearing faults based on two-dimensional convolutional neural networks. *Modeling and Simulation*. **13** 1333. (2024).
- Chiu, P. H. et al. CAN-PINN: A fast physics-informed neural network based on coupled-automatic-numerical differentiation method[J]. *Comput. Methods Appl. Mech. Eng.* **395**, 114909 (2022).
- Shao, J., Wang, R. & Wang, B. Z. Theoretical Analysis of Rotational Symmetry Models for Time-Domain PINN. In: *2024 IEEE International Conference on Computational Electromagnetics (ICCEM)*. (China, 2024).
- Garcia Inda, A. J., Huang, S. Y., Immamoglu, N. & Yu, W. Physics informed neural network (PINN) for noise-robust phase-based magnetic resonance electrical properties tomography. In: *2022 3rd URSI Atlantic and Asia Pacific Radio Science Meeting (AT-AP-RASC)*, (1–4), (Spain, 2022).
- New, A., Eng, B., Timm, A. C. & Gearhart, A. S. Tunable complexity benchmarks for evaluating physics-informed neural networks on coupled ordinary differential equations. In: *2023 57th Annual Conference on Information Sciences and Systems (CISS)*, (1–8). (Baltimore, 2023).
- Liu, F., Li, J. & Wang, L. P. I. L. S. T. M. Physics-informed long short-term memory network for structural response modeling. *Eng. Struct.* <https://doi.org/10.1016/j.engstruct.2023.116500> (2023).
- Wang, L. E. J., Yang, R., Wang, C., Li, H. & Xiong, R. A physics-informed neural network-based method for predicting degradation trajectories and remaining useful life of supercapacitors. *Green. Energy Intell. Transp.* **4** (3), 100291. <https://doi.org/10.1016/j.geits.2025.100291> (2025).
- Gnanasambandam, R., Shen, B., Chung, J., Yue, X. & Kong, Z. Self-scalable Tanh (Stan): multi-scale solutions for physics-informed neural networks. *IEEE Transact. Pattern Analysis Mach. Intell.* **45**, 12, 15588–15603. (2023).
- Lange, M. D. et al. A continual learning survey: defying forgetting in classification tasks. *IEEE Trans. Pattern Anal. Mach. Intell.* **44** (7), 3366–3385 (2022).
- Hadsell, R. et al. Embracing change: continual learning in deep neural networks. *Trends Cogn. Sci.* **24** (12), 1028–1040 (2020).
- Mermillod, M. & Bugaiska, A. P. Bonin the stability-plasticity dilemma: investigating the continuum from catastrophic forgetting to age-limited learning effects. *Front. Psychol.* **4** 504. (2013).
- Oh, H., Han, B., McCluskey, P., Han, C. & Youn, B. D. Physics-of-failure, condition monitoring, and prognostics of insulated gate bipolar transistor modules: A review. *IEEE Trans. Power Electron.* **30**, 2413–2426 (2015).
- Ciappa, M. Selected failure mechanisms of modern power modules. *Microelectron. Reliab.* **42**, 653–667 (2002).
- Norris, K. C. & Landzberg, A. H. Reliability of controlled collapse interconnections. *IBM J. Res. Dev.* **13**, 266–271 (1969).
- Bayerer, R., Herrmann, T., Licht, T., Lutz, J. & Feller, M. Model for power cycling lifetime of IGBT modules—Various factors influencing lifetime. In: *5th International Conference on Integrated Power Electronics Systems* (1–6).
- Abuelnaga, A., Narimani, M. & Bahman, A. S. A review on IGBT module failure modes and lifetime testing. *IEEE Access*. **9**, 9643–9663 (2021).
- Kovacevic, I. F., Drofenik, U. & Kolar, J. W. New physical model for lifetime estimation of power modules. In: *Te 2010 International Power Electronics Conference—ECCE ASIA 2106–2114*.
- Deng, S. et al. A Data-Driven Approach for Power IGBT Operation State Prediction Based on Domain Adapted BiLSTM Networks. In: *2023 International Conference on Sensing, Measurement & Data Analytics in the era of Artificial Intelligence (ICSMD)*, (China, 2023).
- Zhao, S., Peng, Y., Zhang, Y. & Wang, H. Parameter Estimation of power electronic converters with physics-informed machine learning. *IEEE Trans. Power Electron.* **37**, 11567–11578 (2022).
- Zhang, M., Xu, Q. & Wang, X. Physics-informed neural network based online impedance identification of voltage source converters. *IEEE Trans. Ind. Electron.* **70**, 3717–3728 (2023).
- Shen, J. et al. An embedded physical information network for blade crack detection considering dynamic multi-level credibility. *Mech. Syst. Signal Process.* **224**, 111948 (2025).
- Wu, X. et al. AutoPINN: when automl Meets physics-informed neural networks, (2022).
- Chen, S., Zhang, J., Wang, S., Wen, P. & Zhao, S. Circuit parameter identification of degrading DC-DC converters based on physics-informed neural network. In: *2022 Prognostics and Health Management Conference (PHM-2022 London)* 260–268.
- Lu, Y. & Christou, A. Prognostics of IGBT modules based on the approach of particle filtering. *Microelectron. Reliab.* **92**, 96–105 (2019).
- Arias Chao, M., Kulkarni, C., Goebel, K. & Fink, O. Fusing physics-based and deep learning models for prognostics. *Reliab. Eng. Syst. Saf.* **217**, 107961 (2022).
- Li, Z. Z. D. Hoiem learning without forgetting IEEE trans pattern anal Mach Intell, **40** (12) 2935–2947. (2018).

38. Shen Junxian, G. et al. An unsupervised mixed-up differential replay mechanism for propagation detection of blade crack. *Adv. Eng. Inform.* **68**, 103732 (2025).
39. Xu, A. et al. Strategic integration of adaptive sampling and ensemble techniques in federated learning for aircraft engine remaining useful life prediction. *Appl. Soft Comput.* **175**, 113067 (2025).
40. Cai, C. et al. FedCov: Enhanced trustworthy federated learning for machine RUL prediction with continuous-to-discrete conversion. *IEEE Transact. Industrial Inform.* **20**, 11, 12565–12574 (2024). <https://doi.org/10.1109/TII.2024.3423314>
41. Liu, J. Federated learning with prototype learning for RUL prediction of engine turbine blade. *Mech. Syst. Signal Process.* **238**, 113209. <https://doi.org/10.1016/j.ymssp.2025.113209> (2025).
42. Guo, L. et al. FedRUL: A new federated learning method for edge-cloud collaboration based remaining useful life prediction of machines. *IEEE/ASME Transactions on Mechatronics* **28**, 1, 350–359, (2023). <https://doi.org/10.1109/TMECH.2022.3195524>
43. Chen, X., Wang, H., Lu, S. & Yan, R. Bearing remaining useful life prediction using federated learning with Taylor-expansion network pruning. *IEEE Trans. Instrumentation Measur.*, **72**, 1–10, 3512610, (2023). <https://doi.org/10.1109/TIM.2023.3264027>
44. Dhar, P. et al. *Learning without memorizing Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
45. Douillard, A. et al. PODNet: pooled outputs distillation for small-tasks incremental learning comput vis ECCV 86–102. (2020).

Author contributions

F.H. conceived the research idea, developed the Physics-Informed Neural Networks methodology, implemented the continual learning framework, conducted the experiments, analyzed the data, and wrote the main manuscript text. B.M. supervised the research, provided guidance on the theoretical framework, contributed to the experimental design, and reviewed and revised the manuscript. Both authors discussed the results and contributed to the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to F.H.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com